

Single Source Justification (Simplified Acquisition)

Section A: General Contract Information

Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)? ☐ Yes ☒ No

Contracting Activity: PZI (Arnold) OL Holloman

Project/Program Name: Manikin Replacement Parts

Purchase Request / Local ID Number: F2ETGT6036A101

Estimated Contract Cost (including Options): \$21,365.00

Section B: Description of the Supplies/Services Required

This is a request for a Firm Fixed Price purchase order for the acquisition of sole source, proprietary replacement parts required for the repair and sustainment of mission-essential government-owned equipment. The proposed contractor is Humanetics Innovative Solutions, Inc.

The 846th Test Squadron (846 TS) requires specific OEM mechanical replacement parts to repair two manikins, LOIS-9 and LARD-23. These critical Anthropomorphic Test Devices (ATDs) were damaged during mandatory USAF aircraft emergency egress system testing and are currently unserviceable, directly impacting the 846 TS's ability to conduct its mission. The required parts are proprietary components of these custom-made manikin systems, originally designed and manufactured by Humanetics Innovative Solutions, Inc., the same vendor that supplied the complete ATDs under contract FA860624CB002. Only parts from the OEM will ensure the required form, fit, function, and biofidelic certification of the repaired manikins. The specific parts required are detailed in the Statement of Work (SOW), Attachment 1.

Section C: Justification for Soliciting from a Single Source or Brand Name

Justification for this sole-source action is made under the authority of FAR 6.302-1, "Only one responsible source and no other supplies or services will satisfy agency requirements." Humanetics Innovative Solutions, Inc. is the only source capable of providing the required proprietary parts that will satisfy the agency's minimum needs, a fact confirmed in writing by an official sole source letter from the OEM which is documented in the Market Research Report. Specifically, the Government does not own the drawings, manufacturing specifications, intellectual property, or other technical data package (TDP) for the LOIS and LARD manikin systems. These data rights are proprietary to the Original Equipment Manufacturer (OEM), Humanetics Innovative Solutions, Inc. Without the TDP, the government cannot solicit this requirement from any other source, as another manufacturer would be unable to produce the parts to the exact material, dimensional, and biofidelic specifications required.

Furthermore, the required items are not generic hardware; they are integral, certified biofidelic components that mimic human response in high-speed tests. The use of parts from any other manufacturer, even if reverse-engineered, would invalidate the ATD's airworthiness certification for use in egress testing. This would compromise the integrity and validity of all future test data, creating unacceptable safety and performance risks for aircrew life-saving systems and wasting millions of dollars in test funding. A full recertification effort, which would be required if non-OEM parts were used, would be cost-prohibitive and cause an unacceptable mission delay of over a year. Other alternatives were considered and dismissed; reverse-engineering is not feasible due to the small quantity and complex nature of the parts, and acquiring entirely new manikins at an estimated cost of \$600K each is not an economical alternative.

Therefore, purchasing the OEM replacement parts is the only solution that satisfies the government's minimum needs without causing unacceptable schedule delays, cost increases, and risk to the mission.

Section D: Efforts to Obtain Competition

Efforts to obtain competition are not practical for this sustainment action. As documented in the attached Market Research report, mandatory sources were cleared, and no other sources were identified. The acquisition history for these specific systems, including contract number FA860624CB002 for the full ATDs and contract number

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FA910125PB065 for other proprietary components, has consistently established this as a sole-source procurement to the OEM. While other ATD manufacturers exist (e.g., Kistler, CTS), they produce their own proprietary product lines and do not possess the technical data required to manufacture interchangeable parts for Humanetics' custom designed manikins. These research efforts confirm that competition is not available for this requirement.

Section E: Steps to Preclude Future Single Source or Brand Name Awards

For the Current Requirement, no actions can be taken to increase competition for this specific sustainment effort. The cost and schedule impact of reverse-engineering, testing, and certifying a new source for a small quantity of proprietary parts is prohibitive and presents an unacceptable risk to the program.

To preclude future sole-source sustainment, the program office (AFLCMC/ROU) may consider several strategies for new ATD system acquisitions. These include negotiating for Government Purpose Rights (GPR) for critical technical data during the initial contract award or specifying an open, modular system architecture. These actions would enable future competition for replacement parts and upgrades but are not feasible for the sustain-and-repair of the current, existing equipment.

Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(1\)](#), that the circumstances of this contract action deem only a single source is reasonably available.

Date 13-Mar-2026	Typed Name and Rank/Grade of Contracting Officer Tyler S. Payne, Contracting Officer AFTC/PZIH (Arnold), 340-6372	Signature of Contracting Officer PAYNE.TYLER.S.1376784750 Digitally signed by PAYNE.TYLER.S.1376784750 Date: 2026.03.13 14:18:39 -05'00'
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